# Fewer Rounds, Fewer Bytes, Smaller Privacy Budgets: Communication-Budgeted Federated Learning for Clinical Prediction over Constrained Links

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***Abstract*—Cross-silo federated learning (FL) between hospitals is limited less by computation than by the uplink: every synchronisation round is a scheduled, audited transfer over an institutional WAN, a cellular edge slice or a satellite backhaul. We treat the learning job as a network tenant with an explicit communication quota and study, on a patient-deduplicated readmission benchmark with ten seeds per configuration and 1148 runs in total, how the number of rounds in which a site transmits governs three costs at once: accuracy, network cost, and the privacy budget under client-level differential privacy (DP). We introduce TierFed, a scheduler that measures per-tier cross-client dispersion and solves, exactly and in under a millisecond, the integer programme that minimises a local-SGD drift bound subject to the quota, while keeping a per-site classifier head private. At a quota of $\rho = 0.25$, TierFed reaches 0.6269 AUROC against 0.6215 for full-rate FedPer, non-inferior on AUROC and on AUPRC at a 0.01 margin, with 70% less upload; composed with sparse ternary compression and error feedback it uploads 1.6 MB per site instead of 217.8 MB at 0.6290 AUROC. Under an identical client-level DP mechanism with an exact Rényi accountant, transmitting in fewer rounds is charged a smaller budget and injects noise fewer times: at noise multiplier $\sigma = 1$, TierFed reaches 0.6100 AUROC at $\varepsilon = 29.4$ against 0.5669 at $\varepsilon = 67.2$ for FedPer. We also report what does not favour the method. A fixed uniform period and the AdaComm baseline match or exceed TierFed's accuracy at comparable upload without DP; the divergence-driven tier differentiation is rarely active on this data; no method separates from the others under update inversion or membership inference; and a claim of an earlier version of this work, that sparse multi-rate schedules are harmful, does not survive patient-level deduplication and is retracted. Every number in this article is generated from released, bit-reproducible per-seed logs.**

***Index Terms*—Federated learning, communication efficiency, network resource allocation, 6G edge intelligence, differential privacy, gradient inversion, clinical prediction, non-IID data.**

## I. Introduction

FEDERATED learning lets several institutions train one predictive model without exchanging records [1], [2]. In the cross-silo setting that matters for healthcare, the participants are a handful of sites, each behind its own firewall and each connected to the aggregation server by a link that the institution's network team controls [3], [4]. The learning job is then one tenant among many on an uplink with a fixed capacity, a scheduled transfer window and an audit trail. Whether that link is an institutional WAN, a 5G/6G edge slice serving a mobile clinic, or a satellite backhaul serving a remote site, the operator asks how many bytes the job will push per window and whether it will fit [5]–[7].

Prior work reduces communication either by shrinking each message (quantisation, sparsification, sketching [8], [9]) or by sending messages less often (local steps, adaptive periods, lazy uploads [10]–[12]), and a separate line keeps part of the model local for personalisation [13]–[17]. Layer-wise schedules that synchronise different parameter blocks at different frequencies [18] sit between these lines. What is less studied is that the number of transmitting rounds is not only a network quantity. Under client-level differential privacy (DP) [19], every transmitting round is a release that the privacy accountant must charge for and into which noise is injected. A communication quota therefore also sets a privacy cost, and the two can be traded jointly.

This article makes that coupling explicit and measures it. We cast the federated job as a tenant with a quota $\rho$ on the mean fraction of the shared model transmitted per round, and study how accuracy, network cost and DP budget respond when rounds are spent sparingly. We propose TierFed, a scheduler that measures a per-tier dispersion proxy online and solves, exactly, the integer programme that minimises the drift term of a blockwise local-SGD bound subject to $\rho$, while keeping a per-site classifier head private. We compare it with full-rate personalisation (FedPer), drift correction (FedProx, SCAFFOLD), layer-wise adaptive intervals (a FedLAMA-style reimplementation), adaptive-period local SGD (AdaComm), lazy uploads (LAG) and fixed multi-rate schedules, all on one shared architecture table, with ten seeds, Holm-corrected paired tests and one-sided non-inferiority on both AUROC and AUPRC.

**Findings.** On a patient-deduplicated readmission benchmark:

- *Synchronising less does not cost accuracy.* Every sparse schedule we evaluated, including fixed periods and TierFed, matches or slightly exceeds full-rate FedPer; TierFed at $\rho = 0.25$ is non-inferior to FedPer on AUROC and AUPRC with 70% less upload (Section VIII-A). An earlier version of this work reported the opposite for fixed schedules; that result came from patient leakage across splits and a scheduling bug, and we retract it.

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Code, per-seed logs and the scripts that generate every table of this article are available at https://github.com/Abraheem13/tierfed.

- *Timing and size reductions compose.* With sparse ternary compression and error feedback, TierFed uploads 1.6 MB per site instead of FedPer's 217.8 MB without loss of AUROC (Section VIII-C).
- *Fewer rounds buy privacy.* Under an identical per-client clipping DP mechanism and an exact Rényi accountant, round-sparse methods are charged a smaller $\varepsilon$ and retain more utility at the same noise multiplier; TierFed dominates the full-rate methods at every noise level we tested and is level with AdaComm except at the extremes (Section VIII-E).

We are equally explicit about the limits: a well-chosen fixed period and AdaComm reach comparable or better accuracy at comparable upload without DP; TierFed's divergence-driven differentiation between tiers is rarely active on this data; and neither update inversion nor membership inference separates the methods. The contribution is therefore an operator-facing scheduler with an exact budget solver, together with a careful, reproducible measurement of how the communication quota simultaneously sets network cost and privacy cost, rather than an accuracy improvement.

**Contributions.**

- A network-side formulation of cross-silo FL as a tenant with a communication quota, and a blockwise local-SGD bound whose drift term $\sum_g K_g^2 \tau_{\max}^2 (\Gamma_g + \sigma_g^2)$ motivates per-tier intervals (Sections III, V).
- TierFed: an exact, sub-millisecond solver for the budget-constrained interval programme, driven by online dispersion measurements, composable with any compressor (Section IV).
- A measurement of the communication–privacy coupling under a corrected client-level DP mechanism, against round-sparse and full-rate baselines (Section VIII-E).
- Network-level costing on five link profiles and an attack-surface evaluation on the slices each method actually transmits (Sections VIII-D, VIII-E).
- A reporting discipline: bit-identical reruns, patient-grouped splits, a single architecture table, Holm-corrected tests, non-inferiority on AUROC and AUPRC, every number generated from released logs, and an explicit list of configurations in which the method does not help.

## II. Related Work

**Aggregation under heterogeneity.** FedAvg [1] averages all parameters every round. FedProx [20] adds a proximal term; SCAFFOLD [21] corrects client drift with control variates at twice the payload. Surveys catalogue many variants [2], [22]; all synchronise everything every round.

**Personalisation.** FedPer [13] keeps a private classification head; FedRep [14] alternates representation and head updates; LG-FedAvg [15] keeps lower layers local and shares the top; FedBABU [16] trains only the body during federation; Pillutla *et al.* [17] give a general treatment of partial-model personalisation; Ditto [23] and FedBN [24] personalise through regularisation and normalisation statistics. FedPer is the strongest baseline in our experiments and is the reference for every non-inferiority test. None of these methods controls *when* the shared part is sent.

**Communication frequency.** Local SGD with a fixed period is analysed in [10], [25]. AdaComm [11] adapts a single period from the training loss to minimise an error-versus-runtime bound; LAG [12] lets a client skip an upload when its update has changed little since the last one; FedPAQ [26] combines periodic averaging with quantisation. These are the closest prior work to a measurement-driven schedule, and we implement AdaComm and LAG as baselines rather than discussing them away. They differ from TierFed in three respects: they use one period for the whole model (AdaComm) or a per-client rather than per-block decision (LAG), they do not accept an explicit budget, and they do not withhold a personalised head.

**Layer-wise aggregation.** FedLAMA [18] assigns per-layer intervals from measured discrepancy; pFedLA [27] learns per-layer weights with a hypernetwork; L-DAWA [28] weights layers by divergence in self-supervised FL; FedCDA [29] aggregates across rounds. TierFed differs in casting the schedule as a budget-constrained minimisation of the bound term $\sum_g K_g^2 \Gamma_g$, in exposing $\rho$ as the single operator-facing knob, and in excluding the head from aggregation. We report a reimplementation of FedLAMA from its published description and label it as such.

**Compression.** Quantisation, sparsification and sketching reduce message size [8], [9]. They are orthogonal to message timing; Section VIII-C shows the two compose multiplicatively.

**FL over wireless and 6G networks.** The communication cost of FL over wireless links has been studied through joint learning-and-communication design [7], over-the-air computation [30], and surveys of FL in mobile edge networks [5], [6]; 6G roadmaps place distributed learning among native network services [31], [32]. Our contribution is complementary: we do not design the physical layer, we give the application a knob that maps directly onto a slice or window quota and we cost every method on published link profiles.

**Privacy and attack surface.** Client-level DP for FL follows McMahan *et al.* [19] with Rényi accounting [33]. Gradient and update inversion [34], [35] and membership inference [36] measure what an adversary recovers. We evaluate all three on every method, on the slices each method actually transmits.

## III. System Model and Problem Statement

### A. Federation and objective

$N$ sites hold private datasets $\mathcal{D}_k$ of size $n_k$, $n = \sum_k n_k$, $p_k = n_k/n$. The shared objective is $F(\theta) = \sum_k p_k F_k(\theta)$ with $F_k(\theta) = \mathbb{E}_{\xi \sim \mathcal{D}_k}[f(\theta; \xi)]$. In round $r$ each participating site initialises the shared part of its model from the broadcast state and performs one local epoch of $\tau_k = \lceil n_k / B \rceil$ optimiser steps with batch size $B$ (Adam in the main experiments; an SGD arm is in the supplement), then returns the shared slice that is due. Table I collects the notation.

### B. Tier partition

Parameters are partitioned into disjoint tiers $\mathcal{G} = \{\mathrm{s}, \mathrm{m}, \mathrm{f}\}$ (slow, medium, fast). Membership is structural: parameters are ordered by their position in the forward computation graph and

TABLE I
NOTATION.

| | |
|---|---|
| $N$, $n_k$, $p_k$ | sites; local sample count; aggregation weight |
| $\theta$, $\theta_g$, $w_g$ | shared parameters; tier $g$; its share of the shared payload |
| $\theta_f$ | private classifier head (never transmitted) |
| $K_g$, $K_{\max}$, $\tau_g$ | interval of tier $g$; cap; round of its last synchronisation |
| $\Gamma_g$, $\hat{\Gamma}_g$ | gradient divergence of tier $g$ (A3); measured dispersion proxy |
| $\rho$ | communication quota (mean payload fraction after warm-up) |
| $T_w$, $R$, $\tau_k$ | warm-up rounds; rounds; local steps per round at site $k$ |
| $\eta$, $L$, $\sigma_g^2$ | step size; smoothness; per-tier gradient variance |
| $C$, $\sigma$, $(\varepsilon, \delta)$ | DP clipping norm; noise multiplier; privacy budget |
| $\delta_{\mathrm{NI}}$ | non-inferiority margin |

TABLE II
LINK PROFILES USED BY THE NETWORK COST MODEL (3). VALUES ARE REPRESENTATIVE TARGETS, NOT MEASUREMENTS, AND ARE THE ONLY ASSUMPTIONS IN THE NETWORK-LEVEL RESULTS.

| Profile | $\beta$ (Mbit/s) | $\ell$ (ms) | $P$ (W) | $W$ (s) |
|---|---|---|---|---|
| NR cell edge [37] | 5 | 30 | 0.2 | 60 |
| NR mid-cell [37] | 50 | 20 | 0.2 | 60 |
| 6G edge slice (target) [32] | 1000 | 5 | 0.5 | 60 |
| LEO satellite uplink | 10 | 40 | 2.0 | 60 |
| Institutional WAN (wired) | 100 | 10 | 0 | 60 |

cut at depth fractions $\varphi_1 < \varphi_2$, with the classifier head always assigned to the fast tier. For the architectures in this article the model declares the cut explicitly (input projection, hidden blocks, head), which the structural rule reproduces. The fast tier is private, which recovers FedPer-style personalisation as the special case $K_{\mathrm{s}} = K_{\mathrm{m}} = 1$. The parameter counts and tier shares of every model are taken from the model instantiated on the real data and reported in Section VII, Table IV.

### C. Multi-rate aggregation and payload

Let $\mathcal{A}_r \subseteq \mathcal{G}$ be the tiers due in round $r$ and $\mathcal{S}_r$ the participating sites. The server aggregates only the active tiers,

$$\theta_g^{(r+1)} = \begin{cases} \sum_{k\in\mathcal{S}_r} n_k\, \theta_{k,g}^{(r)} \big/ \sum_{k\in\mathcal{S}_r} n_k, & g \in \mathcal{A}_r, \\ \theta_g^{(r)}, & g \notin \mathcal{A}_r, \end{cases} \tag{1}$$

and sites keep their own fast tier throughout. With $w_g$ the share of the shared payload carried by tier $g$, the mean payload fraction over $R$ rounds is

$$\Pi(\{K_g\}) = \frac{1}{R}\sum_{r=1}^{R}\sum_{g\in\mathcal{A}_r} w_g \;\longrightarrow\; \sum_{g\in\mathcal{G}_{\mathrm{sh}}} \frac{w_g}{K_g}. \tag{2}$$

Rounds with $\mathcal{A}_r = \emptyset$ transmit nothing; these local-only rounds are the mechanism by which a quota below $\min_g w_g$ is realised. The realised fraction differs from $\rho$ because warm-up rounds are always full and the head is never sent; both the target and the realised fraction are reported.

### D. Network cost model

The learning job is costed on a link profile $(\beta, \ell, P)$: uplink rate $\beta$ (bit/s), round-trip latency $\ell$, transmit power $P$. Sites upload in parallel on independent links, so a round in which $b_k$ bits are sent by site $k$ costs

$$t_r = \max_k \frac{b_k}{\beta} + \ell, \qquad E_r = \sum_k \frac{b_k}{\beta}\, P, \tag{3}$$

and nothing when no site sends. Communication time and energy are summed over rounds; window compliance is the fraction of rounds whose $t_r$ fits a scheduled window $W$. Table II lists the five profiles, chosen to span a cell-edge 5G NR uplink, a mid-cell NR uplink, a 6G sub-THz edge slice target, an LEO satellite uplink and a shared institutional WAN. The model deliberately omits scheduling, fading and retransmission; it is a cost translation of measured bits under stated assumptions, and every method is costed on the same profiles.

### E. Problem statement

Given a quota $\rho \in (0, 1]$ on the mean shared-payload fraction, choose the synchronisation intervals $\{K_g\}$ each round so that the federated model's accuracy is non-inferior to the full-rate personalised baseline while the payload satisfies (2) and the attack surface, measured as in Section VI, does not grow.

## IV. TierFed: a Budget-Optimal Controller

### A. Divergence measurement

TierFed estimates the heterogeneity of each tier from the dispersion of the slices it actually receives. For tier $g$ with tensors $j \in g$,

$$\hat{\Gamma}_g^{(r)} = \frac{\sum_{j\in g} \frac{1}{|\mathcal{S}_r|} \sum_{k\in\mathcal{S}_r} \big\|\theta_{k,j}^{(r)} - \bar{\theta}_j^{(r)}\big\|_2^2}{\sum_{j\in g} \big\|\bar{\theta}_j^{(r)}\big\|_2^2 + \epsilon}, \tag{4}$$

a scale-free ratio that makes tiers of different magnitude comparable. Two properties of this estimator are stated now so that no claim rests on them implicitly. First, it is available only for tiers received in round $r$; for the others the last estimate is carried forward and its age is logged. Second, because the received slices are the result of $K_g\tau_k$ local steps, $\hat{\Gamma}_g$ scales with the interval that produced it (Section V); it is a proxy for the drift the bound penalises, not an unbiased estimate of $\Gamma_g$.

### B. The integer programme

Given a quota $\rho$, the controller solves

$$\{K_g^\star\} = \arg\min_{K_g\in\{1,\dots,K_{\max}\}} \sum_{g\in\mathcal{G}_{\mathrm{sh}}} K_g^2\, \hat{\Gamma}_g \quad \text{s.t.} \quad \sum_{g\in\mathcal{G}_{\mathrm{sh}}} \frac{w_g}{K_g} \le \rho. \tag{5}$$

With at most three shared tiers and $K_{\max} = 8$ the feasible set has at most $8^3 = 512$ points, so (5) is solved exactly by enumeration in well under a millisecond (Table XX). We also report the greedy doubling rule of an earlier version of this work, which lengthens the interval of the tier with the smallest marginal bound cost per unit of bandwidth saved,

$$g^\star = \arg\min_g \frac{\hat{\Gamma}_g\big[(2K_g)^2 - K_g^2\big]}{w_g/K_g - w_g/(2K_g)} = \arg\min_g \frac{6\,\hat{\Gamma}_g K_g^3}{w_g}, \tag{6}$$

and which reaches only powers of two. The closed form exposes the controller's preference: sparsity goes first to tiers with low divergence, a large payload share and a short current interval.

**Algorithm 1** TierFed: server round $r$

**Require:** quota $\rho$, warm-up $T_w$, cap $K_{\max}$, intervals $\{K_g\}$, last-sync $\{\tau_g\}$
determine $\mathcal{A}_r$ by (7)
**if** $\mathcal{A}_r = \emptyset$ **then**
  **return** {local-only round: zero payload}
**end if**
broadcast $\{\theta_g^{(r)}\}_{g\in\mathcal{A}_r}$ to $\mathcal{S}_r$
**for all** $k \in \mathcal{S}_r$ in parallel **do**
  load broadcast tiers; restore private head $\theta_{f,k}$
  run one local epoch; return (optionally compressed) active tiers
**end for**
(if DP) clip each site's delta to $C$, sum, add $\mathcal{N}(0, \sigma^2 C^2 I)$, divide by $|\mathcal{S}_r|$
aggregate active tiers by (1)
update $\hat{\Gamma}_g$ for $g \in \mathcal{A}_r$ by (4); $\tau_g \leftarrow r$
$\{K_g\} \leftarrow$ exact solution of (5)

### C. Scheduling rule and warm-up

Tier $g$ is due when $r - \tau_g \geq K_g^\star$, where $\tau_g$ is the round of its last synchronisation:

$$\mathcal{A}_r = \begin{cases} \mathcal{G}_{\mathrm{sh}}, & r \leq T_w, \\ \{g :\ r - \tau_g \geq K_g^\star\}, & r > T_w. \end{cases} \tag{7}$$

Divergence is largest in the earliest rounds, so the first $T_w$ rounds synchronise every shared tier. $T_w = 5$ and $K_{\max} = 8$ are held fixed across every dataset, architecture, loss and privacy setting in this article; a $T_w$ sweep is in Table XVII.

### D. Composition with compression

TierFed controls *when* a tier is sent; a compressor controls *how large* the message is. For a compressor with rate $c \in (0, 1]$ bits per parameter relative to float32 the expected upload over $R$ rounds is

$$B = R\,|\mathcal{S}|\,\Pi(\{K_g\})\,c\,32\,d_{\mathrm{sh}} \text{ bits}, \tag{8}$$

with $d_{\mathrm{sh}}$ the shared parameter count; the reduction factors multiply. Section VIII-C tests this on five compressors.

### E. What the operator sets

The only operator-facing quantity is $\rho$. It is a quota on the fraction of the shared model transmitted per round on average, so it maps directly onto a slice allocation or a transfer-window budget through (8) and (3). The paper reports the mapping $\rho \mapsto$ (upload, communication time, energy) on every link profile so that an operator can read off what a given quota buys.

## V. Convergence Analysis

The analysis is for the shared backbone with the private heads frozen at their current values, i.e. $F_k(\theta) = F_k(\theta; \theta_{f,k})$; the personalisation gap this introduces is not bounded and is measured empirically by the FedPer-versus-FedAvg contrast of Table VIII. We make the standard assumptions of local-SGD analysis [10], [25], stated per tier.

**Assumption 1** (A1–A4)**.** *Each $F_k$ is L-smooth; stochastic gradients are unbiased with per-tier variance $\mathbb{E}\|\nabla_g f(\theta;\xi) - \nabla_g F_k(\theta)\|^2 \leq \sigma_g^2$; the per-tier heterogeneity is bounded,*

$$\frac{1}{N}\sum_{k=1}^{N} \left\|\nabla_g F_k(\theta) - \nabla_g F(\theta)\right\|^2 \leq \Gamma_g \quad \forall\theta,\ \forall g; \tag{9}$$

*and $F \geq F^\star$.*

**Proposition 1** (Multi-rate local-SGD bound)**.** *Under A1–A4, with local SGD of step size $\eta$, $\tau_k \in [\tau_{\min}, \tau_{\max}]$ local steps per round, and tier $g$ averaged every $K_g$ rounds, there are absolute constants $c_0, c_1, c_2$ such that for $\eta \leq c_0/(LK_{\max}\tau_{\max})$,*

$$\frac{1}{R}\sum_{r=1}^{R} \mathbb{E}\left\|\nabla F(\bar{\theta}_r)\right\|^2 \leq \underbrace{\frac{c_1\,(F(\bar{\theta}_1) - F^\star)}{\eta\,\tau_{\min} R}}_{\textit{optimisation}} + \underbrace{c_1\,\eta L \sum_g \frac{\sigma_g^2}{N}}_{\textit{stochastic}} + \underbrace{c_2\,\eta^2 L^2 \sum_g K_g^2 \tau_{\max}^2 \left(\Gamma_g + \sigma_g^2\right)}_{\textit{multi-rate drift}}. \tag{10}$$

*Proof sketch.* Apply the perturbed-iterate argument of [25] blockwise. Between two synchronisations of tier $g$ a site takes at most $K_g\tau_{\max}$ local steps on that block, so the squared deviation of its block iterate from the virtual average is bounded by $\eta^2 K_g^2 \tau_{\max}^2(\Gamma_g + \sigma_g^2)$ up to the constants of the cited theorem; the blocks are disjoint, so the deviations add, and the descent lemma with the step-size condition absorbs the cross terms. Setting $K_g = 1$ for a single block recovers the local-SGD rate with $\tau$ local steps. The derivation at the level of the cited theorem is in the supplement. □

**The design rule.** The cost of a long interval is $K_g^2\tau_{\max}^2(\Gamma_g + \sigma_g^2)$: quadratic in the interval, linear in heterogeneity. A tier tolerates sparsity if and only if its divergence is small. Because $\Gamma_g$ depends on the data, the architecture and the training stage, the bound gives no reason to expect one fixed schedule to be right in general; whether this matters in practice is an empirical question, and on the benchmark of Section VIII-A it largely does not.

**What the controller observes.** The controller cannot observe $\Gamma_g$. It observes the dispersion (4) of the slices it receives, which, by the same argument as in the proof, is itself of order $\eta^2 K_g^2 \tau^2(\Gamma_g + \sigma_g^2)$ after $K_g$ rounds of local training. $\hat{\Gamma}_g$ is therefore a noisy, scale-free proxy for the drift term of tier $g$ at its current interval. We log $\hat{\Gamma}_g$ together with the interval that produced it and we do not claim that $\hat{\Gamma}_g$ estimates $\Gamma_g$.

**Optimiser.** The bound is for SGD; the benchmark uses Adam with persistent state for every method because that is the solver the baselines were introduced with and that practitioners use. An SGD arm is reported in Table XI; it shows that the ordering of methods does depend on the local solver.

**What is not claimed.** Nothing above proves that (5) is the right programme, only that its objective is the shape of an upper bound on the drift term. The controller is evaluated

empirically against fixed schedules, FedLAMA-style adaptive intervals, AdaComm and LAG.

## VI. Threat Model and Privacy Mechanisms

### A. Threat model

We consider an honest-but-curious aggregator and a passive eavesdropper on the uplink, both of whom observe exactly the bytes a site transmits in each round: the broadcast state and the site's returned slice of the active tiers. Neither party observes the private head, local data or local optimiser state. We do not consider Byzantine participants or model poisoning. The attack surface of a method is therefore the sequence of slices it transmits, and we measure it in three ways on every method.

### B. Update inversion

The adversary reconstructs a batch $\hat{x}$ from an observed update $\Delta_{\mathcal{A}}$ restricted to the transmitted tiers by solving

$$\hat{x} = \arg\min_{x} \; 1 - \cos\Big(\nabla_{\theta_{\mathcal{A}}} \ell\big(h_\theta(x), y\big), \; -\Delta_{\mathcal{A}}\Big), \qquad (11)$$

the cosine objective of Geiping *et al.* [35] without an image prior, since the inputs are tabular. Unlike a single-batch gradient attack [34], the observed update is the result of a full local epoch of Adam steps, which is the update an aggregator actually sees; this is the update-inversion setting of [35]. The attack is run inside training, on the first site's real transmitted slice, in every round in which that site transmits, for every method, so a method that transmits rarely is attacked on the rounds in which it does. Success is the normalised reconstruction error and the feature correlation between each true row and its best-matching reconstruction; lower correlation means a smaller attack surface. The attack is intentionally generic: the aim is a like-for-like comparison of what each method exposes, not the strongest possible adversary.

### C. Membership inference

Against the deployed (personalised) model of every site we run the loss-threshold attack of Yeom *et al.* [36]: the adversary labels a record a training member when the model's loss on it is low. The attack AUC over members and patient-grouped non-members, averaged over sites, measures memorisation of the released model; $0.5$ is no leakage.

### D. Client-level differential privacy

For client-level DP the server clips every site's returned delta and adds noise to the sum before averaging:

$$\tilde{\Delta} = \frac{1}{|\mathcal{S}_r|}\Big( \sum_{k \in \mathcal{S}_r} \Delta_k \min\big(1, \tfrac{C}{\|\Delta_k\|_2}\big) + \mathcal{N}(0, \sigma^2 C^2 I)\Big), \quad (12)$$

the DP-FedAvg mechanism of McMahan *et al.* [19] with uniform weights, so that replacing one site changes the sum by at most $C$ in $\ell_2$. The mechanism is applied by the server to every method identically. Only rounds in which something is transmitted consume budget; the accountant is the exact Rényi divergence of the Gaussian mechanism, $\alpha/(2\sigma^2)$ per charged round, composed over charged rounds and converted with the standard bound [33],

$$\varepsilon = \min_{\alpha > 1} \Big\{ \frac{R_{\mathrm{tx}}\, \alpha}{2\sigma^2} + \frac{\log(1/\delta)}{\alpha - 1} \Big\}, \qquad \delta = 10^{-5}. \qquad (13)$$

Full participation is used, so no subsampling amplification is claimed. We note that an earlier version of this work clipped the aggregate rather than each site's delta; that mechanism does not bound the sensitivity to one site and its reported budgets were invalid. The budgets reported here are as computed, however large; we do not claim that any of them constitutes meaningful privacy, only that the trade-off is measured correctly.

TABLE III
Dataset characteristics, from the run logs.

| | Diabetes 130-US | PneumoniaMNIST |
|---|---|---|
| Modality | tabular EHR | imaging |
| Train / test rows | 55,992 / 13,998 | 5,232 / 624 |
| Patients (train) | 55,992 | – |
| Features / channels | 160 | 1 ($28 \times 28$) |
| Positive rate (%) | 9.0 | 74.2 |
| Sites | 20 (specialty, capped) | 20 (Dirichlet 0.3) |
| Largest site share (%) | 12 | 14 |
| Label TV (mean) | 0.009 | 0.370 |

## VII. Experimental Design

### A. Data

**Diabetes 130-US Hospitals** [38] is the main tabular benchmark: 30-day readmission from encounter-level features. Cleaning follows Strack *et al.*: encounters discharged to death or hospice are removed and only the first encounter per patient is kept, which leaves 55,992 training and 13,998 test encounters from 55,992 patients, 160 encoded features, and a positive rate of 9.0%. The public release carries no hospital identifier, so sites are formed from the medical-specialty column by longest-processing-time bucketing with a client-share cap of 15%: a specialty larger than the cap, including the "unknown specialty" group that covers roughly half of the encounters, is split across clients, patient-grouped, before bucketing. The largest client therefore holds 12% of the training data and the mean label total-variation distance between clients is 0.009. We call these clients *sites* rather than hospitals. The train/test split and every per-client held-out split are patient-grouped.

**eICU-CRD** [39] carries a true `hospitalid`. The released code includes the loader and the reproduction script runs it when the credentialed files are present; it was not run for this version. **PneumoniaMNIST** [40] is the imaging benchmark: a natively binary $28 \times 28$ collection, partitioned by Dirichlet label skew ($\alpha = 0.3$) with a minimum client size of 50 images. Table III summarises; all dataset statistics in this article are read from the run logs.

### B. Models and tiers

Four backbones share one architecture table used by every method: an MLP (hidden 128–128–64), a tabular ResNet

TABLE IV
AUTOMATIC TIER PARTITION ON THE REAL DATA (FROM THE RUN LOGS). THE FAST TIER IS THE PRIVATE HEAD.

| Architecture | Params | Slow (%) | Medium (%) | Fast (%) |
|---|---|---|---|---|
| MLP (Diabetes) | 45,506 | 45.3 | 54.4 | 0.3 |
| CNN (PneumoniaMNIST) | 287,586 | 3.4 | 96.5 | 0.1 |

(width 128, depth 6), an FT-Transformer (width 64, depth 3, 8 heads) [41] and a small CNN for imaging. Table IV reports the parameter counts and tier shares of each model as instantiated on the real data.

### C. Protocol

$R = 60$ rounds, 20 sites at full participation, one local epoch, batch 64, Adam at $10^{-3}$ with persistent state, inverse-frequency class weighting, $T_w = 5$, $K_{\max} = 8$, $\rho = 0.25$ unless stated. Baselines use their published defaults (FedProx $\mu = 0.01$; FedLAMA-style $\tau = 5$, $\phi = 2$; AdaComm $\tau_0 = 8$; LAG $\xi = 1$); none were tuned on the test data and neither was TierFed. Because personalised methods hold private heads, every method is evaluated on each site's patient-grouped held-out split with that site's own parameters and the per-site metrics are sample-weighted; sites whose held-out split contains a single class cannot yield an AUROC and are excluded identically for every method, with their count reported. A pooled AUROC over all held-out predictions and the global model's AUROC on the untouched test set are reported alongside. Compressors act on each site's update relative to the broadcast with per-client error feedback, and communication is counted from what the compressor emitted for the sites that actually sent. In the fixed multi-rate schedule, as in TierFed, a round in which no tier is due is local-only.

### D. Determinism

Repeated runs of one configuration are bit-identical on the machine that produced the logs (deterministic kernels, single-threaded reductions, seeded loaders); the reproduction script asserts this before any run. An earlier version of this work lacked this guarantee and identical configurations differed by up to 0.006 AUROC, which is larger than the effect it tested.

### E. Statistics

All comparisons are paired across the ten shared seeds. We report Holm-corrected paired $t$-test $p$-values over the family of all method-versus-FedPer contrasts, BCa bootstrap intervals of the paired difference, and the paired standardised difference $d_z$ with the small-sample correction. Parity is never inferred from a non-significant difference. A one-sided non-inferiority test at margin $\delta_{\mathrm{NI}} = 0.01$ is run on *both* final AUROC and final AUPRC,

$$t_{\mathrm{NI}} = \frac{\bar{d} + \delta_{\mathrm{NI}}}{s_d/\sqrt{n}}, \qquad p = 1 - \mathcal{T}_{n-1}(t_{\mathrm{NI}}), \tag{14}$$

and the non-inferiority tests are reported uncorrected and interpreted against the Holm-corrected difference tests. The margin is a reporting convention: 0.01 AUROC is below the between-seed standard deviation of every method in Table V; it has no clinical derivation and we do not claim one. Early-stopped and last-ten-round statistics are reported so that no conclusion depends on the final round alone.

## VIII. RESULTS

Every table body and every number in this section is generated from the per-seed logs by the released analysis script. Unless stated otherwise, values are means over ten seeds on Diabetes 130-US with the MLP backbone; $p$-values are from paired tests across seeds.

### A. Main benchmark

Table V reports the main benchmark. Full-rate FedPer reaches 0.6215 AUROC at 217.8 MB per site. TierFed at $\rho = 0.25$ reaches 0.6269 (paired difference +0.0054, $p = 0.006$) at 65.3 MB, transmitting in 18 of 60 rounds; it is non-inferior to FedPer on AUROC ($p =< 10^{-4}$) and on AUPRC ($p =< 10^{-4}$). The round-sparse baselines are at least as accurate: AdaComm reaches 0.6302 at 29.1 MB (TierFed minus AdaComm: -0.0033, $p = 0.024$), and the fixed multi-rate schedule ($K_{\mathrm{s}} = 5$, $K_{\mathrm{m}} = 2$) reaches 0.6339 at 82.9 MB. LAG at its default threshold never skips an upload and is equivalent to full-rate averaging (0.6351); SCAFFOLD with the Adam local solver used for every method underperforms (0.5930) and is reported with its native SGD solver in Table XI. Early-stopped AUROC is within a narrow band for all methods, so no conclusion depends on the final round.

**Synchronising less does not hurt here.** Table VII sweeps the fixed schedule's periods. A uniform period of four rounds for both shared tiers reaches 0.6407 at 58.1 MB, the best accuracy of any configuration in this study; only a long slow-tier period ($K_{\mathrm{s}} = 8$) costs accuracy (0.6130 with $K_{\mathrm{m}} = 1$). Table VIII separates the private head from the schedule: FedPer and FedAvg are level (0.6215 and 0.6203), so the head does not matter on this data, and TierFed with and without the head is also level (0.6269 and 0.6264). The modest gain of every sparse schedule over full-rate averaging is consistent with additional local training acting as personalisation. An earlier version of this work reported that fixed multi-rate schedules lose about 0.05 AUROC to FedPer; that finding was produced by encounters of the same patient appearing in training and test splits, and by a fallback rule that sent the medium tier whenever no tier was due. Both are corrected here and the finding does not reproduce.

### B. The budget knob and the controller

Table IX shows that $\rho$ behaves as an operator knob: upload rises monotonically with the quota, from 45.7 MB at $\rho = 0.15$ to 167.2 MB at $\rho = 0.8$. Accuracy is flat above $\rho = 0.25$ and drops only at $\rho = 0.15$ (0.6070); the best quota is $\rho = 0.4$ (0.6366 at 84.9 MB). The last column is the honest test of the divergence-driven mechanism: at $\rho = 0.25$ the solver gave the two shared tiers different intervals in 0% of rounds, at $\rho = 0.4$ in 7%, and at most in 26% at $\rho = 0.15$. On this data

TABLE V
MAIN BENCHMARK (DIABETES 130-US, 20 SITES, 60 ROUNDS, 10 SEEDS). FINAL-ROUND FEDERATED AUROC (MEAN ± SD), AUPRC, EARLY-STOPPED AUROC, POOLED AUROC, CUMULATIVE UPLOAD PER SITE, NUMBER OF TRANSMITTING ROUNDS, AND ONE-SIDED NON-INFERIORITY $p$-VALUES AGAINST FEDPER AT $\delta_{\mathrm{NI}} = 0.01$ ON AUROC AND AUPRC.

| Method | $n$ | AUROC | AUPRC | ES-AUROC | Pooled | Upload (MB) | Tx rounds | $\mathrm{NI}_{\mathrm{AUROC}}$ $p$ | $\mathrm{NI}_{\mathrm{AUPRC}}$ $p$ |
|---|---|---|---|---|---|---|---|---|---|
| FedAvg | 10 | 0.6203 ± 0.0100 | 0.171 | 0.6417 | 0.6209 | 218.4 | 60 | $< 10^{-4}$ | $< 10^{-4}$ |
| FedProx | 10 | 0.6279 ± 0.0107 | 0.175 | 0.6433 | 0.6290 | 218.4 | 60 | $< 10^{-4}$ | $< 10^{-4}$ |
| SCAFFOLD | 10 | 0.5930 ± 0.0137 | 0.138 | 0.6119 | 0.5930 | 436.9 | 60 | 0.997 | 1.000 |
| FedPer | 10 | 0.6215 ± 0.0087 | 0.171 | 0.6419 | 0.6201 | 217.8 | 60 | (ref.) | (ref.) |
| FedLAMA | 10 | 0.6291 ± 0.0101 | 0.176 | 0.6421 | 0.6295 | 154.1 | 60 | $< 10^{-4}$ | $< 10^{-4}$ |
| AdaComm | 10 | 0.6302 ± 0.0071 | 0.168 | 0.6365 | 0.6290 | 29.1 | 8 | $< 10^{-4}$ | $< 0.001$ |
| LAG | 10 | 0.6351 ± 0.0124 | 0.179 | 0.6407 | 0.6362 | 218.4 | 60 | $< 10^{-4}$ | $< 10^{-4}$ |
| Fixed schedule | 10 | 0.6339 ± 0.0100 | 0.177 | 0.6395 | 0.6338 | 82.9 | 37 | $< 10^{-4}$ | $< 10^{-4}$ |
| TierFed | 10 | 0.6269 ± 0.0082 | 0.175 | 0.6415 | 0.6261 | 65.3 | 18 | $< 10^{-4}$ | $< 10^{-4}$ |

TABLE VI
HOLM-CORRECTED PAIRED COMPARISONS ON FINAL AUROC AGAINST FEDPER (10 SEEDS): MEAN DIFFERENCE, BCA 95% INTERVAL, CORRECTED $p$, PAIRED $d_z$.

| Comparison | Δ | 95% CI | $p_{\mathrm{Holm}}$ | $d_z$ | Sig. |
|---|---|---|---|---|---|
| AdaComm vs FedPer | +0.0087 | [0.0070, 0.0105] | $< 0.001$ | 1.66 | ✓ |
| FedAvg vs FedPer | -0.0012 | [-0.0038, 0.0004] | 0.578 | -0.33 | |
| FedLAMA vs FedPer | +0.0076 | [0.0058, 0.0101] | 0.001 | 1.91 | ✓ |
| FedProx vs FedPer | +0.0065 | [0.0042, 0.0085] | 0.005 | 1.57 | ✓ |
| Fixed schedule vs FedPer | +0.0124 | [0.0101, 0.0156] | $< 0.001$ | 2.44 | ✓ |
| LAG vs FedPer | +0.0136 | [0.0100, 0.0166] | $< 0.001$ | 2.23 | ✓ |
| SCAFFOLD vs FedPer | -0.0284 | [-0.0397, -0.0196] | 0.005 | -1.57 | ✓ |
| TierFed vs FedPer | +0.0054 | [0.0023, 0.0079] | 0.036 | 1.03 | ✓ |

TABLE VII
FIXED-SCHEDULE PERIOD SWEEP $(K_{\mathrm{s}}, K_{\mathrm{m}})$, 10 SEEDS; ROUNDS WITH NO TIER DUE ARE LOCAL-ONLY.

| $(K_{\mathrm{s}}, K_{\mathrm{m}})$ | $n$ | AUROC | Upload (MB) |
|---|---|---|---|
| (2, 1) | 10 | 0.6317 ± 0.0097 | 170.0 |
| (2, 2) | 10 | 0.6362 ± 0.0110 | 112.5 |
| (4, 1) | 10 | 0.6314 ± 0.0102 | 145.3 |
| (4, 2) | 10 | 0.6370 ± 0.0112 | 87.8 |
| (4, 4) | 10 | 0.6407 ± 0.0114 | 58.1 |
| (8, 1) | 10 | 0.6130 ± 0.0079 | 132.1 |
| (8, 2) | 10 | 0.6155 ± 0.0077 | 74.6 |
| (8, 4) | 10 | 0.6254 ± 0.0073 | 44.9 |
| (8, 8) | 10 | 0.6250 ± 0.0088 | 29.0 |

the controller mostly reduces to a uniform period chosen to meet the quota, which explains why a well-chosen fixed period performs as well. Where the solvers differ, exactness helps: at $\rho = 0.4$ the exact solver reaches 0.6366 at 84.9 MB against 0.6283 at 95.7 MB for the greedy doubling rule (Table X).

Table XI gives the baselines a fair frontier. AdaComm with $\tau_0 = 8$ is the most byte-efficient configuration without DP (0.6302 at 29.1 MB); LAG skips uploads only at $\xi = 10^4$, where its accuracy collapses (0.5673). With an SGD local solver SCAFFOLD recovers (0.6383), FedAvg and FedPer improve (0.6396, 0.6379), and TierFed is lower (0.6251); the ordering of methods therefore depends on the local solver, which we state as a limitation.

TABLE VIII
ATTRIBUTION (10 SEEDS): PRIVATE HEAD × SCHEDULE, AND TIERFED WITH AND WITHOUT THE PRIVATE HEAD.

| Cell | $n$ | AUROC | AUPRC | Upload (MB) |
|---|---|---|---|---|
| fedavg | 10 | 0.6203 ± 0.0100 | 0.171 | 218.4 |
| fedper | 10 | 0.6215 ± 0.0087 | 0.171 | 217.8 |
| fixed | 10 | 0.6339 ± 0.0100 | 0.177 | 82.9 |
| sched_only | 10 | 0.6336 ± 0.0102 | 0.177 | 83.5 |
| tierfed | 10 | 0.6269 ± 0.0082 | 0.175 | 65.3 |
| tierfed_nohead | 10 | 0.6264 ± 0.0073 | 0.174 | 65.5 |

TABLE IX
QUOTA SWEEP FOR TIERFED (10 SEEDS): ACCURACY, UPLOAD, REALISED PAYLOAD FRACTION, MEAN INTERVALS CHOSEN BY THE EXACT SOLVER, AND THE SHARE OF POST-WARM-UP ROUNDS IN WHICH THE TWO SHARED TIERS RECEIVED DIFFERENT INTERVALS.

| $\rho$ | $n$ | AUROC | AUPRC | Upload (MB) | Realised | $\bar{K}_{\mathrm{s}}$ | $\bar{K}_{\mathrm{m}}$ | Differentiated (%) |
|---|---|---|---|---|---|---|---|---|
| 0.15 | 10 | 0.6070 ± 0.0119 | 0.163 | 45.7 | 0.21 | – | – | – |
| 0.25 | 10 | 0.6269 ± 0.0082 | 0.175 | 65.3 | 0.30 | – | – | – |
| 0.40 | 10 | 0.6366 ± 0.0119 | 0.179 | 84.9 | 0.39 | – | – | – |
| 0.60 | 10 | 0.6290 ± 0.0093 | 0.176 | 116.2 | 0.53 | – | – | – |
| 0.80 | 10 | 0.6296 ± 0.0094 | 0.176 | 167.2 | 0.77 | – | – | – |

### C. Scheduling and compression compose

Table XII confirms (8): reductions from timing and from message size multiply. 8-bit quantisation alone takes FedPer from 217.8 to 54.5 MB at unchanged accuracy (0.6213); TierFed with 8-bit quantisation uploads 16.4 MB (0.6266); and TierFed with sparse ternary compression uploads 1.6 MB at 0.6290, against 5.2 MB (0.6250) for FedPer with the same compressor. Count-sketch compression fails for every method (0.4986, 0.5017, 0.5049): at the sketch width used, the unsketched estimate of the update is too noisy for this model, and we report it rather than tune it away.

### D. Network-level cost

Table XIII translates measured bits into what an operator sees. On a cell-edge NR uplink the total communication time of the job falls from 19.2 s for FedPer to 5.8 s for TierFed. The model is small, so every method fits the 60-s window on every link profile; the absolute times matter less than the proportional reductions, which carry over to larger models with the same schedule.

TABLE X
EXACT SOLVER VERSUS GREEDY DOUBLING AT TWO QUOTAS (10 SEEDS).

| $\rho$ / solver | $n$ | AUROC | Upload (MB) |
|---|---|---|---|
| 0.15 / exact | 10 | 0.6070 ± 0.0119 | 45.7 |
| 0.25 / exact | 10 | 0.6269 ± 0.0082 | 65.3 |
| 0.25 / greedy | 10 | 0.6269 ± 0.0082 | 65.3 |
| 0.40 / exact | 10 | 0.6366 ± 0.0119 | 84.9 |
| 0.40 / greedy | 10 | 0.6283 ± 0.0112 | 95.7 |
| 0.60 / exact | 10 | 0.6290 ± 0.0093 | 116.2 |
| 0.80 / exact | 10 | 0.6296 ± 0.0094 | 167.2 |

TABLE XI
BASELINE SWEEPS (10 SEEDS): ADACOMM INITIAL PERIOD $\tau_0$, LAG THRESHOLD $\xi$, AND THE SGD LOCAL SOLVER FOR FEDAVG, FEDPER, SCAFFOLD AND TIERFED.

| Configuration | $n$ | AUROC | AUPRC | Upload (MB) | Tx rounds |
|---|---|---|---|---|---|
| adacomm tau16 | 10 | 0.6125 ± 0.0111 | 0.152 | 18.2 | 5 |
| adacomm tau2 | 10 | 0.6301 ± 0.0095 | 0.177 | 109.2 | 30 |
| adacomm tau4 | 10 | 0.6275 ± 0.0068 | 0.174 | 54.6 | 15 |
| adacomm tau8 | 10 | 0.6302 ± 0.0071 | 0.168 | 29.1 | 8 |
| fedavg sgd | 10 | 0.6396 ± 0.0109 | 0.172 | 218.4 | 60 |
| fedper sgd | 10 | 0.6379 ± 0.0110 | 0.170 | 217.8 | 60 |
| lag xi10 | 10 | 0.6351 ± 0.0124 | 0.179 | 218.4 | 60 |
| lag xi100 | 10 | 0.6351 ± 0.0124 | 0.179 | 218.4 | 60 |
| lag xi1000 | 10 | 0.6352 ± 0.0126 | 0.179 | 168.4 | 46 |
| lag xi10000 | 10 | 0.5673 ± 0.0498 | 0.135 | 6.8 | 2 |
| scaffold sgd | 10 | 0.6383 ± 0.0106 | 0.173 | 436.9 | 60 |
| tierfed sgd | 10 | 0.6251 ± 0.0095 | 0.157 | 65.3 | 18 |

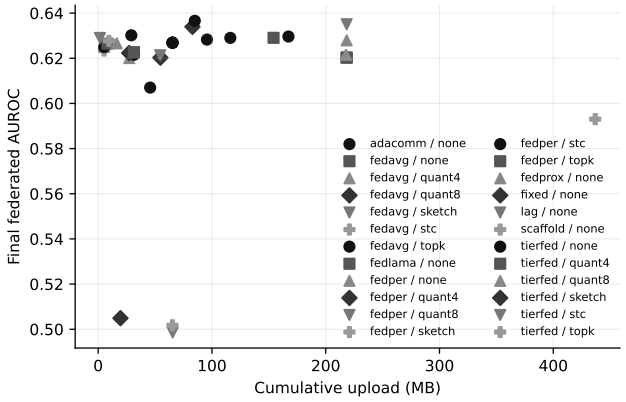



Fig. 1. Accuracy–communication frontier on Diabetes (10 seeds): every method, quota and compressor on one pair of axes.

TABLE XII
SCHEDULING × COMPRESSION (10 SEEDS). COMPRESSORS ACT ON THE UPDATE WITH PER-CLIENT ERROR FEEDBACK. TIERFED ROWS USE $\rho = 0.25$.

| Method / compressor | $n$ | AUROC | AUPRC | Upload (MB) |
|---|---|---|---|---|
| FedAvg / none | 10 | 0.6203 ± 0.0100 | 0.171 | 218.4 |
| FedAvg / quant4 | 10 | 0.6201 ± 0.0094 | 0.172 | 27.4 |
| FedAvg / quant8 | 10 | 0.6203 ± 0.0101 | 0.171 | 54.7 |
| FedAvg / sketch | 10 | 0.4986 ± 0.0075 | 0.090 | 65.5 |
| FedAvg / stc | 10 | 0.6234 ± 0.0113 | 0.172 | 5.2 |
| FedAvg / topk | 10 | 0.6216 ± 0.0107 | 0.171 | 31.5 |
| FedPer / none | 10 | 0.6215 ± 0.0087 | 0.171 | 217.8 |
| FedPer / quant4 | 10 | 0.6223 ± 0.0097 | 0.171 | 27.3 |
| FedPer / quant8 | 10 | 0.6213 ± 0.0085 | 0.172 | 54.5 |
| FedPer / sketch | 10 | 0.5017 ± 0.0043 | 0.091 | 65.3 |
| FedPer / stc | 10 | 0.6250 ± 0.0099 | 0.173 | 5.2 |
| FedPer / topk | 10 | 0.6227 ± 0.0092 | 0.172 | 31.4 |
| TierFed / none | 10 | 0.6269 ± 0.0082 | 0.175 | 65.3 |
| TierFed / quant4 | 10 | 0.6265 ± 0.0091 | 0.174 | 8.2 |
| TierFed / quant8 | 10 | 0.6266 ± 0.0083 | 0.174 | 16.4 |
| TierFed / sketch | 10 | 0.5049 ± 0.0058 | 0.092 | 19.6 |
| TierFed / stc | 10 | 0.6290 ± 0.0087 | 0.174 | 1.6 |
| TierFed / topk | 10 | 0.6277 ± 0.0087 | 0.174 | 9.4 |

### *E. Attack surface and differential privacy*

Table XIV and Fig. 2 contain the clearest effect in this study. Under client-level DP, every transmitting round is a release that the accountant charges for and into which noise is injected, so the number of transmitting rounds sets both the budget and the cumulative noise. At $\sigma = 1$, full-rate FedAvg collapses to 0.5008 and FedPer reaches 0.5669, both at $\varepsilon = 67.2$; TierFed reaches 0.6100 at $\varepsilon = 29.4$ (TierFed minus FedPer +0.043, $p =< 10^{-4}$). At $\sigma = 2$ the margin over FedPer is +0.045 ($p =< 10^{-4}$) at less than half the budget. Against AdaComm, the strongest round-sparse baseline, the comparison is mixed and we report it as such: level at $\sigma = 0.5$ (+0.001, $p = 0.604$) and $\sigma = 1$ (+0.002, $p = 0.443$), better at $\sigma = 2$ (+0.018, $p =< 10^{-4}$), and worse at $\sigma = 4$ (-0.018, $p = 0.008$), where AdaComm's eight releases incur less noise than TierFed's eighteen. The operational reading is that, under DP, the number of rounds a site transmits is the dominant design variable, and a communication quota is also a privacy quota. None of the budgets reported is claimed to constitute meaningful privacy; they are reported as computed so that the trade-off is measured correctly.

Without DP, neither attack separates the methods (Table XV). Update-inversion feature correlation is low for every method (TierFed 0.089, FedPer 0.088, AdaComm 0.065), and membership-inference AUC is close to chance (TierFed 0.540, FedPer 0.535). Per transmitted round, TierFed exposes the same slice as FedPer; its reduction in attack surface is only that it transmits in fewer rounds, which this generic attack does not exploit.

### *F. Robustness, architectures and imaging*

TierFed matches or exceeds FedPer under every class-imbalance loss (Table XVI; e.g. LDAM 0.6276 versus 0.6150). A longer warm-up buys a small gain at a proportional cost in bytes (Table XVII; $T_w = 20$: 0.6354 at 108.9 MB). On the six-block tabular ResNet, TierFed reaches 0.6279 at 603.5 MB against 0.6137 at 2006.0 MB for FedPer, and on the FT-Transformer 0.6345 at 165.2 MB against 0.6352 at 581.2 MB (Table XVIII). As on the MLP, the fixed multi-rate schedule is at least as accurate as TierFed on every backbone (ResNet 0.6382 at 1006.8 MB; FT-Transformer 0.6386 at 270.7 MB), while TierFed uploads less. The FT-Transformer rows other than the fixed schedule were trained on GPU with the fused attention kernel, which is not bit-deterministic; they are reported over five seeds with that caveat.

On PneumoniaMNIST (Table XIX) the picture differs from the tabular benchmark and we report it as it is. The two methods that share the classifier head are the most accurate, FedAvg

TABLE XIII
NETWORK COST OF THE MAIN BENCHMARK UNDER THE LINK PROFILES OF TABLE II: TOTAL COMMUNICATION TIME ON CELL-EDGE NR, LEO SATELLITE AND THE 6G EDGE-SLICE TARGET, UPLINK ENERGY ON CELL-EDGE NR, AND THE FRACTION OF ROUNDS FITTING A 60-S WINDOW ON CELL-EDGE NR. MEANS OVER 10 SEEDS, COMPUTED FROM MEASURED BITS BY (3).

| Method | Upload (MB) | $t$ NR-edge (s) | $t$ LEO (s) | $t$ 6G (s) | $E$ NR-edge (J) | Window (%) |
|---|---|---|---|---|---|---|
| FedAvg | 218.4 | 19 | 11 | 0.4 | 69.9 | 100 |
| FedProx | 218.4 | 19 | 11 | 0.4 | 69.9 | 100 |
| SCAFFOLD | 436.9 | 37 | 20 | 0.5 | 139.8 | 100 |
| FedPer | 217.8 | 19 | 11 | 0.4 | 69.7 | 100 |
| FedLAMA | 154.1 | 14 | 9 | 0.4 | 49.3 | 100 |
| AdaComm | 29.1 | 3 | 1 | 0.1 | 9.3 | 100 |
| LAG | 218.4 | 19 | 11 | 0.4 | 69.9 | 100 |
| Fixed schedule | 82.9 | 8 | 5 | 0.2 | 26.5 | 100 |
| TierFed | 65.3 | 6 | 3 | 0.1 | 20.9 | 100 |

TABLE XIV
UTILITY UNDER CLIENT-LEVEL DP (PER-CLIENT CLIPPING $C = 1$, 10 SEEDS): $(\varepsilon, \delta = 10^{-5})$ FROM THE EXACT RÉNYI ACCOUNTANT, NUMBER OF CHARGED (TRANSMITTING) ROUNDS, FINAL AUROC AND AUPRC. THE SAME MECHANISM IS APPLIED TO EVERY METHOD.

| Method / $\sigma$ | $n$ | $\varepsilon$ | Charged rounds | AUROC | AUPRC |
|---|---|---|---|---|---|
| AdaComm / 0.5 | 10 | 43.2 | 8 | 0.6223 ± 0.0085 | 0.158 |
| AdaComm / 1.0 | 10 | 17.6 | 8 | 0.6076 ± 0.0115 | 0.148 |
| AdaComm / 2.0 | 10 | 7.8 | 8 | 0.5469 ± 0.0135 | 0.123 |
| AdaComm / 4.0 | 10 | 3.6 | 8 | 0.5460 ± 0.0159 | 0.108 |
| FedAvg / 0.5 | 10 | 194.4 | 60 | 0.5990 ± 0.0156 | 0.147 |
| FedAvg / 1.0 | 10 | 67.2 | 60 | 0.5008 ± 0.0096 | 0.096 |
| FedAvg / 2.0 | 10 | 26.1 | 60 | 0.5143 ± 0.0227 | 0.099 |
| FedAvg / 4.0 | 10 | 11.2 | 60 | 0.5150 ± 0.0151 | 0.094 |
| FedPer / 0.5 | 10 | 194.4 | 60 | 0.6250 ± 0.0081 | 0.161 |
| FedPer / 1.0 | 10 | 67.2 | 60 | 0.5669 ± 0.0136 | 0.133 |
| FedPer / 2.0 | 10 | 26.1 | 60 | 0.5196 ± 0.0151 | 0.111 |
| FedPer / 4.0 | 10 | 11.2 | 60 | 0.5198 ± 0.0101 | 0.109 |
| Fixed schedule / 0.5 | 10 | 132.4 | 37 | 0.6248 ± 0.0107 | 0.156 |
| Fixed schedule / 1.0 | 10 | 47.7 | 37 | 0.5923 ± 0.0163 | 0.142 |
| Fixed schedule / 2.0 | 10 | 19.2 | 37 | 0.5372 ± 0.0185 | 0.118 |
| Fixed schedule / 4.0 | 10 | 8.5 | 37 | 0.5155 ± 0.0167 | 0.109 |
| TierFed / 0.5 | 10 | 76.8 | 18 | 0.6230 ± 0.0100 | 0.160 |
| TierFed / 1.0 | 10 | 29.4 | 18 | 0.6100 ± 0.0095 | 0.151 |
| TierFed / 2.0 | 10 | 12.4 | 18 | 0.5644 ± 0.0141 | 0.121 |
| TierFed / 4.0 | 10 | 5.7 | 18 | 0.5282 ± 0.0094 | 0.114 |

(0.9959) and the FedLAMA-style schedule (0.9965); TierFed reaches 0.9698 at 364.1 MB against 0.9643 at 1213.7 MB for FedPer, and is not non-inferior to FedPer at the 0.01 margin ($p = 0.182$). Under strong label skew the private head is a liability rather than an asset: both head-private methods trail the head-sharing ones and FedPer has a large between-seed spread. A schedule that keeps the head private should therefore not be used for this kind of task without a shared head.

### G. Scalability

Table XX reports measured wall-clock and accuracy from 10 to 100 sites with the Dirichlet partition, since the specialty partition cannot produce 100 sites. Upload per site scales linearly with the number of sites for every method and TierFed keeps its proportional saving at every size; seconds per round are dominated by local training and are similar across methods. Accuracy at 100 sites is lower for TierFed than for FedAvg, consistent with each site holding less data. The controller's exact solve enumerates at most 512 points and takes well under a millisecond at every size.

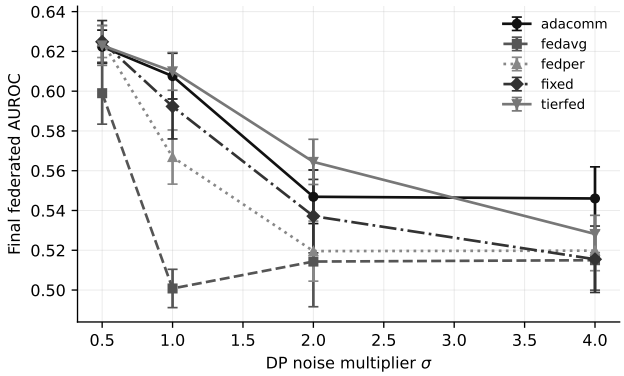



Fig. 2. Utility against the DP noise multiplier for every method under the identical mechanism (12).

TABLE XV
ATTACK SURFACE (10 SEEDS): PARAMETERS EXPOSED PER TRANSMITTING ROUND, UPDATE-INVERSION NORMALISED ERROR AND FEATURE CORRELATION ON THE REAL TRANSMITTED SLICES, MEMBERSHIP-INFERENCE AUC ON THE DEPLOYED MODELS (0.5 = NO LEAKAGE), AND UPLOAD.

| Method | $n$ | Params exposed | NMSE ↑ | Corr. ↓ | MIA AUC ↓ | Upload (MB) |
|---|---|---|---|---|---|---|
| AdaComm | 10 | 45,506 | 11.58 | 0.065 | 0.534 | 29.1 |
| FedAvg | 10 | 45,506 | 9.04 | 0.103 | 0.534 | 218.4 |
| FedPer | 10 | 45,376 | 9.04 | 0.088 | 0.535 | 217.8 |
| Fixed schedule | 10 | 27,509 | 11.93 | 0.048 | 0.526 | 82.9 |
| TierFed | 10 | 45,376 | 10.45 | 0.089 | 0.540 | 65.3 |

## IX. DISCUSSION

**What the evidence supports.** Three claims are supported by ten-seed paired tests on patient-deduplicated data. First, on this benchmark synchronising less does not cost accuracy; every sparse schedule matches or slightly exceeds full-rate personalisation, and TierFed is non-inferior to FedPer on AUROC and AUPRC at a fraction of the upload. Second, timing and message-size reductions compose multiplicatively when compressors act on the update with error feedback, giving more than two orders of magnitude less upload at unchanged accuracy. Third, under client-level DP the number of transmitting rounds is the dominant design variable: round-sparse methods are charged a smaller budget and inject noise fewer times, and they retain substantially more utility at the same noise multiplier. A communication quota is therefore also a privacy quota, which matters operationally because the two are usually negotiated with different stakeholders.

**What it does not support.** TierFed is not more accurate than simpler round-sparse alternatives. A fixed uniform period of four rounds is the most accurate configuration we ran, and AdaComm reaches comparable accuracy with fewer bytes when DP is off. The divergence-driven mechanism that motivates per-tier intervals rarely activates on this data; at most quotas the exact solver returns the same interval for both shared tiers. Under DP, TierFed and AdaComm are level at moderate noise, TierFed is better at $\sigma = 2$, and AdaComm is better at $\sigma = 4$. The case for TierFed over these alternatives is operational rather than statistical: it exposes a quota that maps directly onto a slice or window allocation, solves for the schedule exactly, and composes with compression and DP

TABLE XVI
LOCAL OBJECTIVE SWEEP, FEDPER VERSUS TIERFED ($\rho = 0.25$), 10 SEEDS.

| Loss / method | $n$ | AUROC | AUPRC | Upload (MB) |
|---|---|---|---|---|
| Class-balanced / FedPer | 10 | 0.6232 ± 0.0108 | 0.173 | 217.8 |
| Class-balanced / TierFed | 10 | 0.6264 ± 0.0071 | 0.173 | 65.3 |
| Focal / FedPer | 10 | 0.6198 ± 0.0100 | 0.170 | 217.8 |
| Focal / TierFed | 10 | 0.6237 ± 0.0089 | 0.173 | 65.3 |
| LDAM / FedPer | 10 | 0.6150 ± 0.0114 | 0.164 | 217.8 |
| LDAM / TierFed | 10 | 0.6276 ± 0.0091 | 0.171 | 65.3 |
| Inverse-frequency CE / FedPer | 10 | 0.6215 ± 0.0087 | 0.171 | 217.8 |
| Inverse-frequency CE / TierFed | 10 | 0.6269 ± 0.0082 | 0.175 | 65.3 |

TABLE XVII
WARM-UP SWEEP FOR TIERFED ($\rho = 0.25$), 10 SEEDS.

| $T_w$ | $n$ | AUROC | AUPRC | Upload (MB) |
|---|---|---|---|---|
| 0 | 10 | 0.6270 ± 0.0080 | 0.173 | 54.5 |
| 2 | 10 | 0.6323 ± 0.0092 | 0.176 | 58.1 |
| 5 | 10 | 0.6269 ± 0.0082 | 0.175 | 65.3 |
| 10 | 10 | 0.6289 ± 0.0093 | 0.176 | 79.9 |
| 20 | 10 | 0.6354 ± 0.0112 | 0.179 | 108.9 |

TABLE XVIII
ARCHITECTURES ON DIABETES (5 SEEDS), EVERY METHOD ON THE SAME ARCHITECTURE WITHIN A ROW GROUP; PARAMETER COUNT FROM THE INSTANTIATED MODEL.

| Backbone / method | $n$ | AUROC | AUPRC | Upload (MB) | Params |
|---|---|---|---|---|---|
| mlp / FedAvg | 5 | 0.6236 ± 0.0103 | 0.171 | 218.4 | 0 |
| mlp / FedPer | 5 | 0.6240 ± 0.0087 | 0.172 | 217.8 | 0 |
| mlp / Fixed schedule | 5 | 0.6373 ± 0.0117 | 0.176 | 82.9 | 0 |
| mlp / TierFed | 5 | 0.6298 ± 0.0128 | 0.175 | 65.3 | 0 |
| resnet / FedAvg | 5 | 0.6258 ± 0.0192 | 0.168 | 2007.3 | 0 |
| resnet / FedPer | 5 | 0.6137 ± 0.0107 | 0.161 | 2006.0 | 0 |
| resnet / Fixed schedule | 5 | 0.6382 ± 0.0150 | 0.177 | 1006.8 | 0 |
| resnet / TierFed | 5 | 0.6279 ± 0.0130 | 0.175 | 603.5 | 0 |
| transformer / FedAvg | 5 | 0.6401 ± 0.0134 | 0.177 | 581.8 | 0 |
| transformer / FedPer | 5 | 0.6352 ± 0.0115 | 0.173 | 581.2 | 0 |
| transformer / Fixed schedule | 5 | 0.6386 ± 0.0116 | 0.172 | 270.7 | 0 |
| transformer / TierFed | 5 | 0.6345 ± 0.0083 | 0.171 | 165.2 | 0 |

TABLE XIX
PNEUMONIAMNIST (CNN, 20 SITES, DIRICHLET $\alpha = 0.3$, 10 SEEDS). COLUMNS AS IN TABLE V.

| Method | $n$ | AUROC | AUPRC | ES-AUROC | Pooled | Upload (MB) | Tx rounds | $\text{NI}_{\text{AUROC}}$ $p$ | $\text{NI}_{\text{AUPRC}}$ $p$ |
|---|---|---|---|---|---|---|---|---|---|
| FedAvg | 10 | 0.9959 ± 0.0023 | 0.991 | 0.9957 | 0.9724 | 1214.8 | 60 | 0.037 | $< 0.001$ |
| FedPer | 10 | 0.9643 ± 0.0663 | 0.986 | 0.9790 | 0.9948 | 1213.7 | 60 | (ref.) | (ref.) |
| FedLAMA | 10 | 0.9965 ± 0.0018 | 0.995 | 0.9960 | 0.9837 | 877.4 | 60 | 0.036 | 0.005 |
| Fixed schedule | 10 | 0.9642 ± 0.0677 | 0.989 | 0.9871 | 0.9928 | 614.8 | 37 | $< 10^{-4}$ | 0.002 |
| TierFed | 10 | 0.9698 ± 0.0238 | 0.984 | 0.9866 | 0.9669 | 364.1 | 18 | 0.182 | 0.090 |

accounting, whereas AdaComm exposes a loss-driven period and a fixed schedule exposes nothing an operator can set from a byte budget.

**A retraction.** An earlier version of this work reported that fixed multi-rate schedules lose about 0.05 AUROC to personalised baselines. With one encounter per patient, patient-grouped splits and the fallback-tier bug removed, that result does not reproduce. Readers who relied on it should not.

**Limitations.** (i) The sites of the tabular benchmark are specialty buckets with a share cap, not hospitals; a benchmark with true hospital identifiers (eICU) is supported by the released code but was not run for this version because it requires credentialed access. (ii) The bound is for SGD and the main benchmark uses Adam; the SGD arm shows that the ordering of methods depends on the local solver. (iii) $\hat{\Gamma}_g$ is a dispersion proxy that scales with the interval that produced it; the controller is a feedback loop on that proxy whose stability we observe but do not prove. (iv) The network model omits scheduling, fading and retransmission, and its profiles are representative targets rather than measurements. (v) The inversion attack is generic, the DP budgets are large, and neither should be read as a privacy guarantee. (vi) Full participation is assumed; partial participation would add a sampling term to (10) and permit subsampling amplification in (13). (vii) The FT-Transformer arm was trained with a non-deterministic GPU attention kernel; all other results are bit-reproducible. (viii) On the imaging benchmark under strong label skew, the methods that share the classifier head are the most accurate and both head-private methods trail them; the results here do not transfer to that setting. (ix) Ten seeds on one public tabular dataset and one imaging collection is a benchmark, not a clinical validation.

**Practical guidance.** For a tabular model of this size, an operator who needs a byte budget should set $\rho$ from the transfer window through (8) and (3), compose the schedule with 8-bit quantisation or sparse ternary compression with error feedback, and, when client-level DP is required, prefer the sparsest schedule that meets the accuracy target, because each transmitting round is charged against the privacy budget.

## X. CONCLUSION

We treated a cross-silo federated job as a network tenant with an explicit communication quota and measured, on bit-reproducible runs over patient-deduplicated data, how the number of transmitting rounds sets accuracy, network cost and privacy cost together. Synchronising less did not cost accuracy; timing and size reductions composed to more than two orders of magnitude less upload; and under client-level DP the round count was the dominant variable for both the privacy budget and utility. TierFed turns a byte quota into an exact per-tier schedule and is non-inferior to full-rate personalisation at a fraction of the upload, but it is not more accurate than a well-chosen fixed period or AdaComm, and its divergence-driven differentiation rarely activates on this data. We report these limits, and retract a finding of an earlier version, so that the operational result, that a communication quota is also a privacy quota, rests on evidence a reader can rerun.

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Generative artificial-intelligence tools were used to draft figure layouts and to refactor analysis code; the author reviewed, edited and verified all scientific content, code, numbers and figures. An earlier version of this work was submitted elsewhere under the title “Nested Federated Learning”; it is superseded by this article, which corrects reporting and code defects of that version and retracts its central negative result (Supplement, Section S1).

## DATA AND CODE AVAILABILITY

Diabetes 130-US Hospitals is public (UCI id 296). PneumoniaMNIST is public (MedMNIST v2). eICU-CRD requires

TABLE XX
SCALABILITY (DIABETES, DIRICHLET PARTITION, 30 ROUNDS, 3 SEEDS): ACCURACY, UPLOAD PER SITE, SECONDS PER ROUND AND PEAK GPU MEMORY, MEASURED, FOR 10 TO 100 SITES.

| Sites / method | $n$ | AUROC | Upload (MB) | s/round | Peak mem (MB) |
|---|---|---|---|---|---|
| 10 / FedAvg | 3 | 0.6127 ± 0.0206 | 54.6 | 3.07 | 80 |
| 10 / FedLAMA | 3 | 0.6056 ± 0.0117 | 43.1 | 3.03 | 80 |
| 10 / FedPer | 3 | 0.5935 ± 0.0072 | 54.5 | 3.03 | 80 |
| 10 / TierFed | 3 | 0.6118 ± 0.0100 | 20.0 | 3.01 | 80 |
| 20 / FedAvg | 3 | 0.6187 ± 0.0110 | 109.2 | 3.12 | 87 |
| 20 / FedLAMA | 3 | 0.6187 ± 0.0183 | 85.4 | 3.09 | 87 |
| 20 / FedPer | 3 | 0.6189 ± 0.0235 | 108.9 | 3.11 | 87 |
| 20 / TierFed | 3 | 0.6205 ± 0.0242 | 39.9 | 3.04 | 87 |
| 50 / FedAvg | 3 | 0.6128 ± 0.0224 | 273.0 | 3.40 | 108 |
| 50 / FedLAMA | 3 | 0.6108 ± 0.0203 | 219.3 | 3.33 | 108 |
| 50 / FedPer | 3 | 0.6188 ± 0.0240 | 272.3 | 3.37 | 108 |
| 50 / TierFed | 3 | 0.6048 ± 0.0170 | 99.8 | 3.21 | 108 |
| 100 / FedAvg | 3 | 0.6068 ± 0.0210 | 546.1 | 3.99 | 142 |
| 100 / FedLAMA | 3 | 0.6061 ± 0.0230 | 437.1 | 3.86 | 142 |
| 100 / FedPer | 3 | 0.5929 ± 0.0255 | 544.5 | 3.86 | 142 |
| 100 / TierFed | 3 | 0.5883 ± 0.0157 | 199.7 | 3.57 | 142 |

credentialed PhysioNet access. Code, per-seed logs, the scripts that generate every table and figure, and the determinism check are at https://github.com/Abraheem13/tierfed.

## ETHICS, FUNDING AND COMPETING INTERESTS

All datasets are de-identified and publicly available under their respective licences; no human-subject data were collected and no ethics approval was required. The author declares no competing interests and no external funding.

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